

Srinivasan Poonkundran

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EDUCATION

The University of Arizona

Masters in Science, Data Science

Tucson, AZ

Jan. 2024 – Dec. 2025

Savitribai Phule Pune University

Bachelor of Engineering, Computer

Pune, India

Aug. 2017 – May 2021

EXPERIENCE

Cognizant Technology Solutions, India

Full Stack Engineer

Sept. 2021 – Jan. 2024

- Optimized application load times by 25% using caching strategies and efficient SQL query design.
- Redesigned UI components, improving user engagement by 15% and reducing bounce rates.
- Introduced a new feature that raised customer satisfaction by 20% based on direct user feedback.
- Built a Python-based PDF label validator, saving \$1600 annually in manual effort.
- Developed responsive web interfaces using Angular, HTML, CSS, and TypeScript.
- Built backend microservices with Java and Spring Boot; optimized deployments using AWS services.
- Wrote and optimized complex SQL queries to support application logic and reporting needs.
- Worked in Agile teams, contributing to sprint planning, development, and issue resolution.
- Mentored junior developers and contributed to improving team collaboration and code quality.
- Handled production issues and provided quick technical support to maintain operational stability.
- **Skills** Java, Spring Boot, Angular, TypeScript, SQL, Python, AWS, HTML, CSS, Agile, REST APIs

PROJECTS

Emotion Detection from Text | [GitHub](#)

- Built a BERT-based multi-label emotion classifier to detect 7 emotions from English text.
- Used `bert-base-uncased` for embeddings, frozen during training to reduce resource usage.
- Designed a Keras model with Bidirectional GRUs, dropout, and dense layers for classification.
- Trained with mixed precision on GPU using TensorFlow's ModelCheckpoint and EarlyStopping.
- Preprocessed data with Hugging Face libraries and tuned hyperparameters using micro F1-score.
- Created an end-to-end inference pipeline for prediction and result formatting.
- **Skills** BERT, Hugging Face, TensorFlow, Keras, GRU, Emotion Classification, NLP, Python

Heart Rate Monitoring System | [GitHub](#)

- Built an IoT-enabled web app to monitor heart rate and blood oxygen levels in real-time.
- Used Node.js, Express, and MongoDB to develop the backend with secure RESTful APIs.
- Created a responsive React frontend to display daily and weekly charts from sensor data.
- Integrated MAX30102 sensor with a Photon board using synchronous state machines.
- Enabled configurable time intervals for data capture and ensured offline storage with sync-on-reconnect.
- **Skills** IoT, Node.js, Express, MongoDB, React, MAX30102, Photon, REST API, Real-Time Charts

Cricket Match Result Forecasting | [GitHub](#) | [Website](#)

- Built a Random Forest Classifier to predict ODI match outcomes with 87% accuracy.
- Created a real-time dashboard updating predictions every 5 minutes using simulated match data.
- Developed backend APIs to serve dynamic match stats from historical datasets.
- Improved user engagement with interactive and evolving match outcome predictions.
- **Skills** Random Forest, Python, Dashboard Development, API Design, Cricket Analytics, Real-Time Prediction

Cricket Metrics | [GitHub](#) | [Website](#)

- Developed a dynamic cricket analytics dashboard using 150,000+ historical data points.
- Simulated live match data with backend APIs and integrated with a Quarto-based frontend.
- Delivered insights using line, bar, donut, and map plots for scoring, performance, and trends.

- Built APIs for batting, bowling, and partnerships to support live match analysis and updates.
- **Skills** R, Shiny, Quarto, API Development, Data Visualization, ggplot2, Cricket Analytics

Psychometric Analysis | [GitHub](#) | [Website](#)

- Analyzed data on 890 fictional characters across 100 universes using the Open-Source Psychometrics Project.
- Merged traits, MBTI types, and psychometric scores to study personality and character notability.
- Performed data cleaning, feature engineering, and correlation analysis to uncover key personality links.
- Visualized results using bar charts, density plots, and radar charts to highlight audience engagement trends.
- **Skills** Python, Data Analysis, Psychometrics, MBTI, Data Visualization, Matplotlib, Seaborn

COVID-19 Protocol Monitoring System | [Published in IJRASET](#)

- Built a surveillance system with mask detection, distance monitoring, and temperature sensing using CCTV and microcontrollers.
- Trained a CNN model with 99% accuracy for mask detection using a dataset of 3,500 images.
- Designed a socket-based client-server architecture for multi-node data communication.
- Used Flask and SQLite to deploy a lightweight frontend with reliable backend storage.
- **Skills** Python, CNN, Flask, SQLite, Raspberry Pi Pico, Arduino Uno, OpenCV, Socket Programming

TECHNICAL SKILLS

Programming Languages: Python, R, Java, C++, C, JavaScript, SQL

Data Science & Machine Learning: NumPy, pandas, scikit-learn, XGBoost, Matplotlib, seaborn, Keras, TensorFlow, StatsModels

Development Practices: Object-Oriented Programming, Functional Programming, Microservice Architecture, MVC Pattern, Agile, Test-Driven Development, CI/CD

Databases & Querying: MySQL, SQLite, PostgreSQL, MongoDB, NoSQL

Cloud & DevOps: AWS, Docker

Natural Language Processing: spaCy, NLTK, Transformers (Hugging Face), fastcoref, Gensim, CoreNLP

Web & Backend Technologies: Angular, Spring Boot, Spring MVC, Node.js, Express.js, REST APIs, JWT, Maven, HTML, CSS

Tools & IDEs: Jupyter Notebook, PyCharm, VS Code, RStudio, IntelliJ, Git, GitHub, Postman, Docker Hub

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly

Certifications: Amazon Web Services Certified Cloud Practitioner | [Credentials](#)

COURSEWORK

Core Data Science: INFO 511 – Foundations of Data Science, INFO 523 – Data Mining and Discovery, INFO 502 – Data Ethics

Machine Learning and NLP: INFO 557 – Neural Networks, INFO 539 – Statistical Natural Language Processing, INFO 555 – Applied Natural Language Processing

Data Engineering and Visualization: INFO 526 – Data Analysis and Visualization, INFO 531 – Data Warehousing in the Cloud

Web Development: ECE 513 – Web Development and the Internet of Things